

SensiSpace

Output & Results

Comparison Parameters: ResNet-18 vs Baby Dragon Hatchling

Results at a Glance:

Parameter	ResNet-18	BDH	Winner	Margin
Accuracy	81.98%	95.87%	BDH	+13.89%
Precision	81.64%	95.78%	BDH	+14.14%
Recall	81.51%	96.01%	BDH	+14.50%
F1-Score	81.51%	95.85%	BDH	+14.34%
Parameters	11.18M	2.03M	BDH	5.5x less
Inference	0.18 ms	0.60 ms	ResNet	3.3x fast
Training	245 s	3,658 s	ResNet	15x fast

BDH wins on 5 out of 7 parameters (all quality metrics + model size).

1. Accuracy (Overall Correctness)

"Out of all images, how many did the model classify correctly?"

Model	Accuracy
ResNet-18	81.98%
Baby Dragon Hatchling	95.87%

Formula

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$$

What This Means

Out of 5,400 test images:

- **ResNet-18**: Correctly classified 4,427 images, got 973 wrong
- **BDH**: Correctly classified 5,177 images, got only 223 wrong

Why BDH is Higher

BDH's multi-scale processing captures objects at all sizes (small buildings AND large forests), while ResNet's uniform 3x3 kernels struggle with scale variation in satellite imagery. The lateral inhibition module also sharpens land type boundaries, reducing misclassifications at borders.

2. Precision (Quality of Predictions)

"When the model says 'this is Forest', how often is it actually Forest?"

Model	Precision
ResNet-18	81.64%
Baby Dragon Hatchling	95.78%

Formula

$$\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$$

What This Means

- **ResNet**: When it says "Forest", correct ~82% of the time - 18% are false alarms
- **BDH**: When it says "Forest", correct ~96% of the time - only 4% false alarms

Why It Matters

High precision means fewer false alarms. If you are monitoring deforestation, you don't want the system incorrectly flagging residential areas as forest. Every false alarm wastes investigation resources.

3. Recall (Completeness of Detection)

"Out of all actual Forest images, how many did the model find?"

Model	Recall
ResNet-18	81.51%
Baby Dragon Hatchling	96.01%

Formula

$$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$

What This Means

- **ResNet**: Misses about 18.5% of actual instances of each class
- **BDH**: Misses only about 4% - it finds almost everything

Why It Matters

High recall means fewer missed detections. If you are tracking rivers for flood monitoring, you need the system to find ALL rivers, not miss some. A missed river could mean an undetected flood risk.

4. F1-Score (Balanced Measure)

"A single number that balances precision and recall."

Model	F1-Score
ResNet-18	81.51%
Baby Dragon Hatchling	95.85%

Formula

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

What This Means

F1-Score is the harmonic mean of precision and recall. It is the most reliable single metric because:

- An F1 of 95.85% means BDH has BOTH high precision AND high recall
- You cannot get a high F1 by cheating (e.g., predicting everything as one class)
- Accuracy alone can be misleading with imbalanced datasets; F1 catches this

5. Model Parameters (Complexity)

"How many learnable numbers does the model contain?"

Model	Parameters	Size
ResNet-18	11,181,642	11.18M
Baby Dragon Hatchling	2,032,275	2.03M

ResNet has 5.5x more parameters than BDH.

What This Means

- More parameters = more memory, more computation, harder to train
- BDH achieves higher accuracy with 5.5x fewer parameters
- This proves that intelligent architecture design is more effective than simply having more parameters

Why It Matters for Deployment

- **Cheaper hardware:** Smaller model runs on edge devices, drones, even satellites
- **Less energy:** Lower compute = lower power consumption
- **Faster updates:** Smaller model files transfer and load faster
- **Less overfitting:** Fewer parameters means the model generalizes better

6. Inference Time (Speed)

"How fast can the model classify one image?"

Model	Inference Time
ResNet-18	0.18 ms
Baby Dragon Hatchling	0.60 ms

What This Means

- ResNet is ~3.3x faster per image (simpler architecture, optimized by PyTorch)
- BDH at 0.6ms is still extremely fast - can process ~1,666 images/second
- Both models are well within real-time requirements

Why ResNet is Faster

ResNet uses simple, uniform 3x3 convolutions that are highly optimized in hardware (GPU/MPS). BDH's parallel multi-scale convolutions (3x3 + 5x5 + 7x7), attention modules, and recurrent refinement (2 iterations) add computational overhead - but the 14% accuracy gain is worth the 0.42ms trade-off.

Trade-off Analysis

BDH is 3.3x slower but 14% more accurate. For satellite imagery where accuracy matters more than milliseconds (images are analyzed in batches, not video streams), BDH is the clear winner.

7. Training Time

"How long did it take to train the model?"

Model	Training Time
ResNet-18	245 seconds (~4 min)
Baby Dragon Hatchling	3,658 seconds (~61 min)

BDH takes ~15x longer to train.

Why BDH Takes Longer

- ResNet starts from pretrained ImageNet weights (transfer learning) - already knows basic features
- BDH is trained from scratch - must learn everything from zero
- BDH has recurrent refinement (2 iterations = 2x computation per forward pass)
- BDH has multi-scale branches (3 parallel convolutions per stage)

Why This is Acceptable

Training is a one-time cost. You train once, then deploy. The 61 minutes for BDH is negligible compared to the permanent accuracy advantage. In production, only inference time matters - and both models are real-time capable.

8. Why BDH Outperforms - Module Contribution

Each of the 4 biologically-inspired modules in BDH contributes to a specific quality improvement:

BDH Module	What It Improves	Impact
Multi-Scale Pyramid	Accuracy	Captures small houses AND large forests
Lateral Inhibition	Precision (fewer FP)	Sharpens boundaries between land types
Contextual Attention	Recall (fewer FN)	Focuses on informative features
Recurrent Refinement	F1-Score (overall)	Iteratively cleans up features

9. Conclusion

We evaluated both models using 7 comparison parameters. Baby Dragon Hatchling wins on all 5 quality metrics - accuracy, precision, recall, F1-score, and model size. ResNet-18 wins only on speed metrics (inference time and training time), but both models are fast enough for real-time use.

Parameter	ResNet-18	BDH	Winner	Margin
Accuracy	81.98%	95.87%	BDH	+13.89%
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The key result: 95.87% accuracy with 2.03 million parameters vs 81.98% with 11.18 million - a 14 percentage point improvement with 5.5x fewer parameters. This conclusively demonstrates that biologically-inspired, domain-specific architecture design outperforms generic deep networks for satellite remote sensing.

How to Explain During Demo

"We compare both models on 7 parameters. BDH wins on all 5 quality metrics - accuracy, precision, recall, F1-score, and model size. ResNet is faster, but both are real-time capable. The key result: 95.87% accuracy with 2 million parameters vs 81.98% with 11 million - proving that intelligent architecture design beats brute-force scaling."